Explanation via Surplus Structure

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- what is surplus structure in a scientific theory?
- structural ockham razor (SOCK): posit only those structures that are needed to explain the phenomena
- converse SOCK: if a mathematical structure is needed to explain the phenomena, then it represents something real

quantum field theory
the problem of inequivalent representations

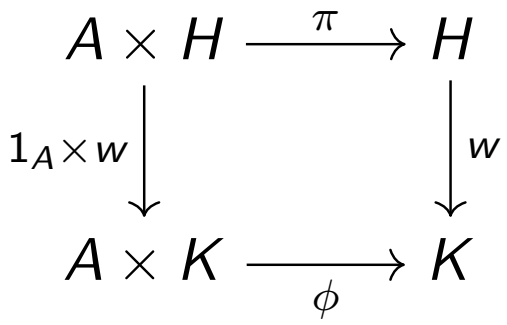
OBJECTIVE: describe a quantum system corresponding to a classical system with infinitely many degrees of freedom $q_1, p_1, q_2, p_2, \ldots$

OBSTACLE: embarrassment of riches — many different, apparently non-equivalent, quantum descriptions

inequivalent representations

- quantization: given some equation $\mathbb{E}(x_1, \ldots, x_n)$, find a Hilbert space H and operators $a_1, \ldots, a_n$ on H that satisfy $\mathbb{E}$.
- algebrizing: there is an abstract algebra A with operators $a_1, \ldots, a_n \in A$ such that:
  1. $E(a_1, \ldots, a_n)$,
  2. representations $\pi : A \to B(H)$ are in one-to-one correspondence with quantizations of $\mathbb{E}$.

unitary equivalence



proposed solutions of INEQ

1. hilbert space conservatism: there is one correct representation
   $\Longrightarrow$ choosing a representation involves a further theoretical commitment

2. algebraic imperialism: all physical information is in the abstract algebra of observables

   $\Longrightarrow$ choosing a representation entails no further theoretical commitment

crash course in algebraic quantum theory

- Observables are represented by operators in A.
- States are defined as mappings from observables to $\mathbb{R}$.
- A representation $\pi : A \to H$ induces a map $x \mapsto \pi^*(x)$ from state vectors in H to states on $\mathcal{A}$.

$$\pi^*(x)(a) \;=\; \langle x, \pi(a)x \rangle.$$

- GNS theorem: for every state ω of A, there is a Hilbert space H, a vector $x \in H$, and a representation $\pi : A \to H$ such that

$$\omega = \pi^*(x).$$

problems for h.s. conservatism

- underdetermination: what reason for choosing a representation?
- what is gained by choosing a representation?
- what is lost by choosing a representation?
- what kind of epistemic commitment is it to choose a representation?
  "The system is in some state $\pi^*(x)$."

problems for algebraic imperialism

- can we do quantum theory without a Hilbert space?
  states = functions from observable algebra to numbers
  dynamics = automorphism of the observable algebra
  parochial observables ???
- too much structure?
  there are too many states

the weak equivalence gambit

- Fell's theorem: for state $x \in H$, observables $a_1, \ldots, a_n \in A$, and $\epsilon > 0$, there is a state $y \in K$ such that

$$|\langle x, a_i x \rangle - \langle y, a_i y \rangle| < \epsilon.$$

- Claim (Segal, Robinson): any two representations are empirically equivalent

don't deflate representation

- Ruetsche: physical equivalence $\Longrightarrow$ unitary equivalence
- Fact: empirical equivalence $\Longrightarrow$ unitary equivalence

do deflate representation

1. intertranslatability $\Longrightarrow$ theoretical equivalence (converse of Glymour 1970)
2. Fact: there are ineq representations that are intertranslatable
3. Conclusion: there are ineq representations that are theoretically equivalent

free Majorana field

- gauge transformation: $\gamma : F \to F$
- observables are fixed points in F under action of γ
- two inequivalent representations: π_+ vacuum, π_- charged
- localized automorphisms of A

$$\pi_-(a) \cong \pi_+(\theta(a)).$$

- H_+ and H_- are dynamical islands
- field operators map H_+ to H_-.

physical equivalence

- The structure of π_+ is isomorphic to the structure of π_-.
- Baker & Halvorson: representations π_+ and π_- are intertranslatable.

physical inequivalence

- π_+ and π_- are unitarily inequivalent.
- The vacuum state Ω is in π_+ and not in π_-.
- In fact, π_+ and π_- are empirically inequivalent.

the Buridan's ass of QFT

1. The world is represented either by π_+ or by π_-, and not by both.
2. If the world is represented by π_+, it could equally well be represented by π_-.

the dilemma restated

1. break the symmetry between the two representations
2. deny that there is any choice to be made

common theme: "freedom to choose a representation" is theoretically insignificant

relations between representations

Majorana particle is its own antiparticle
$\Longrightarrow$ two Majorana particle state is in the vacuum sector

$$\pi_- \otimes \pi_- \cong \pi_+$$

$$\pi_+ \otimes \pi_+ \cong \pi_+$$

conclusions

1. the best explanation might require non-representational structures
2. corollary: essential $\not\Longrightarrow$ representational
3. for future research: what kinds of non-representational structures occur in our best theories, and what theoretical roles do they play?